

IS-004 Sentinel Evidence Engine

Implementation Specification IS-004 - Sentinel Evidence Engine

Status: Approved for Design

Priority: Critical

Package ID: PKG-004

Depends on: IS-001, IS-002, IS-003

Referenced Blueprint Volumes: 22, 24, 56

Vision

The Evidence Engine is the foundation of trust in Project Sentinel. Every conclusion, recommendation, alert, mission, report, AI response and trust score must be backed by evidence.

Core Principle

Evidence is immutable. Interpretation is not.

Evidence Examples

Running process, network connection, certificate, Windows Event Log, Linux syslog entry, browser extension, installed application, file hash, DNS response, registry key, authentication event, cloud API response, vulnerability scan result, AI analysis result and user confirmation.

Evidence Structure

Evidence ID, Source, Collection Time, Collector, Object Reference, Integrity Status, Classification, Confidence, Chain of Custody, Metadata.

Lifecycle

Collected -> Validated -> Stored -> Correlated -> Referenced -> Archived -> Retained / Purged (Policy)

Requirements

REQ-EVIDENCE-001: Every evidence item shall have a globally unique Evidence ID.

REQ-EVIDENCE-002: Evidence shall be immutable after validation.

REQ-EVIDENCE-003: Evidence shall record its source and collection time.

REQ-EVIDENCE-004: Derived evidence shall reference originating evidence.

REQ-EVIDENCE-005: Evidence integrity shall be verifiable.

REQ-EVIDENCE-006: Evidence shall support correlation with other objects.

REQ-EVIDENCE-007: AI-generated evidence shall always be identified as AI-derived.